

Enterprise Knowledge Copilot — Open Source Tools & Libraries

Project: Enterprise Knowledge Copilot
Philosophy: 100% open-source stack — zero vendor lock-in, zero cloud dependencies, complete data sovereignty.

Core Technology Stack

Library / Tool	Version	Role in Architecture	Why This One?
Ollama	Latest	Local LLM Runtime	Hosts LLaMA 3.2 (3B) entirely offline. No GPU required. Runs on standard corporate hardware with 8GB RAM.
LangGraph	0.2+	Agent Orchestration	State machine approach enables deterministic routing, conditional branching (escalation on 0 hits), and multi-turn memory — superior to simple chain-based architectures.
ChromaDB	0.4+	Vector Database	Zero-config, file-based, no server process needed. Perfect for local deployment. Supports metadata filtering with \$and operators for SQL-like queries.
rank-bm25	0.2+	Keyword Search (BM25Okapi)	Industry-standard sparse retrieval algorithm. Excels at exact-match queries (error codes, ticket IDs) where dense vectors fail. Pure Python, no external dependencies.
NetworkX	3.0+	Knowledge Graph	Pure Python directed graph library. Enables relational traversal (system → owner lookups) without requiring a graph database server like Neo4j.
spaCy	3.7+	PII Redaction (NER)	Offline Named Entity Recognition using en_core_web_sm model. Detects PERSON entities for name redaction. Fast inference, no cloud API calls.
SentenceTransformers	2.0+	Embeddings + Cross-Encoder	Provides both the embedding model (BAAI/bge-small-en-v1.5, 33MB) for vectorization and the Cross-Encoder (ms-marco-MiniLM-L-6-v2) for precision reranking.
Streamlit	1.30+	Frontend UI	Rapid prototyping framework for data apps. Provides chat interface, session state

			management, custom CSS injection, and sidebar dashboards with minimal code.
NumPy	1.24+	Vector Mathematics	Cosine similarity computation, embedding array operations, and score normalization for RRF fusion and evaluation metrics.

Embedding & Reranking Models

Model	Source	Parameters	Size	Purpose
BAAI/bge-small-en-v1.5	HuggingFace	33M	33 MB	Document and query embedding. Top-tier performance for its size on MTEB benchmark. Runs entirely on CPU.
ms-marco-MiniLM-L-6-v2	HuggingFace	22M	22 MB	Cross-Encoder reranker. Trained on MS MARCO passage ranking dataset. Processes (query, document) pairs together through a transformer for maximum relevance precision.
LLaMA 3.2	Meta via Ollama	3B	~2 GB	Local language model for answer generation. Chosen as optimal balance between quality and CPU/RAM requirements.
en_core_web_sm	spaCy	—	12 MB	NER model for detecting PERSON, ORG, PRODUCT, GPE entities. Used in PII redaction and entity extraction pipelines.

Why 100% Open Source?

1. **Data Sovereignty:** Enterprise data containing employee PII, system credentials, and internal procedures must never leave the organization's infrastructure. Every component runs locally.
2. **Zero Vendor Lock-in:** No dependency on OpenAI, Anthropic, Google, or any commercial API. The entire stack can be audited, forked, and modified.
3. **Reproducibility:** Open weights and open code mean the system behaves identically across deployments. No API version changes or model deprecations.
4. **Cost:** Zero ongoing API costs. After initial setup, the system runs indefinitely on local hardware at zero marginal cost per query.